Digital Electronics Assignment: Number Systems

1. Convert the following decimal numbers to binary: (a) 25 (b) 47 (c) 128

2. Convert the following binary numbers to decimal: (a) 10101 (b) 110011 (c) 1000001

3. Convert the following binary numbers to hexadecimal: (a) 11010110 (b) 11110000

4. Convert the following hexadecimal numbers to binary: (a) 3A (b) F7

5. Perform binary addition: (a) 1011 + 1101 (b) 1110 + 1011

6. Perform binary subtraction using 2’s complement: (a) 1010 - 0111 (b) 1100 - 1010

7. Represent -25 in 8-bit 2’s complement form.

8. Convert the following octal numbers to decimal: (a) 17 (b) 45

9. Convert the following decimal numbers to octal: (a) 64 (b) 100

10. Explain the difference between 1’s complement and 2’s complement with examples.

11. Perform binary multiplication: (a) 101 × 11 (b) 110 × 10

12. What is BCD? Convert decimal 59 into BCD representation.

13. Convert Gray code 1011 to binary.

14. Convert binary 1101 to Gray code.

15. Explain the importance of number systems in digital electronics.